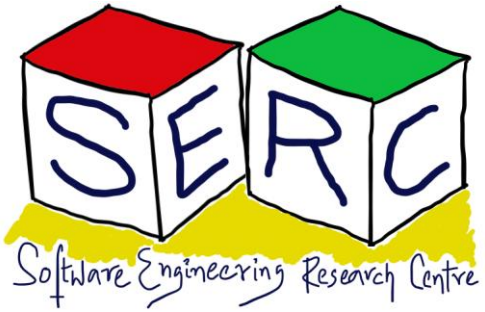
Database Architectures

Software System Development

Monsoon 2026

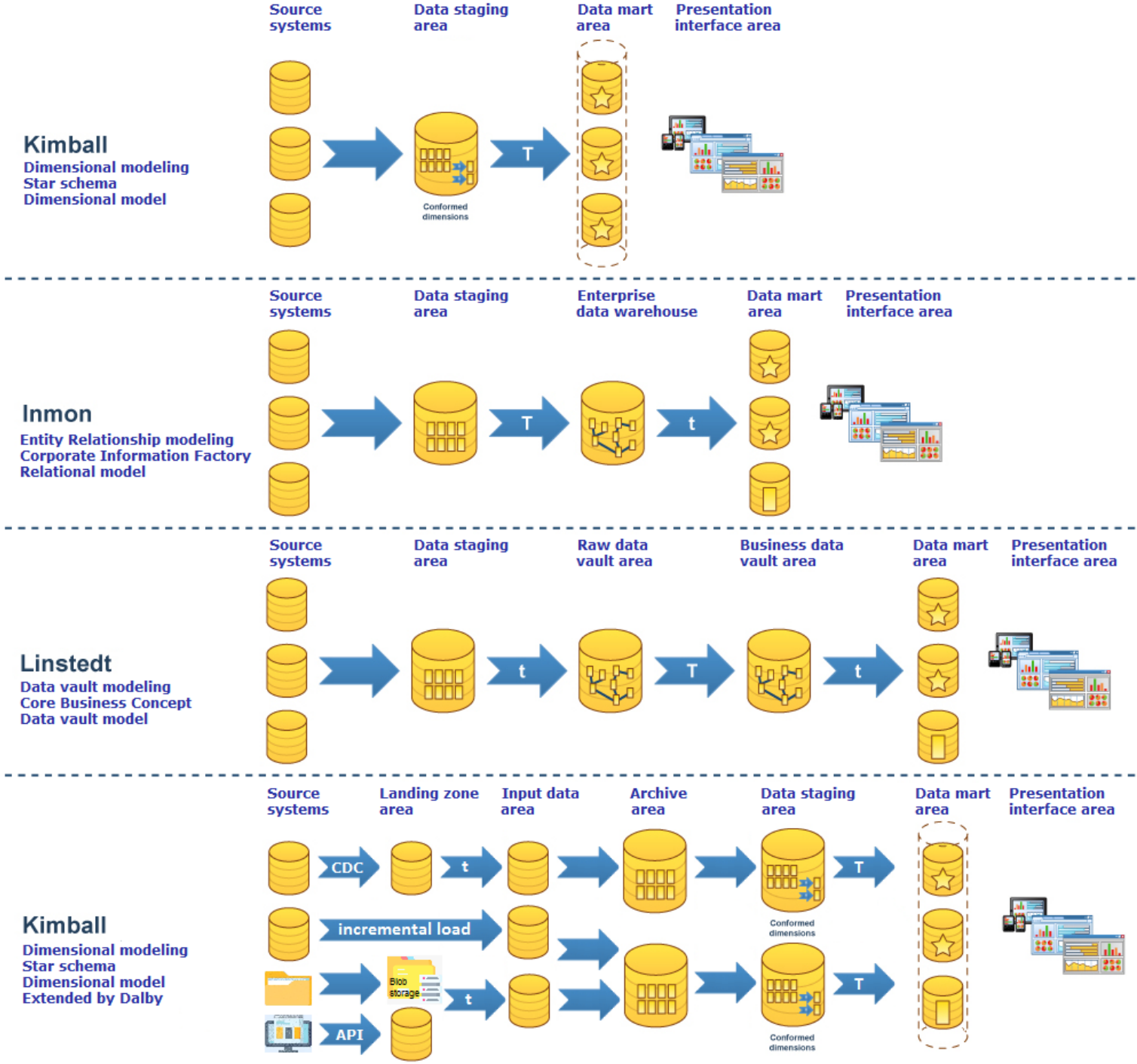
INTERNATIONAL INSTITUTE OF INFORMATION TECHNOLOGY

HYDERABAD



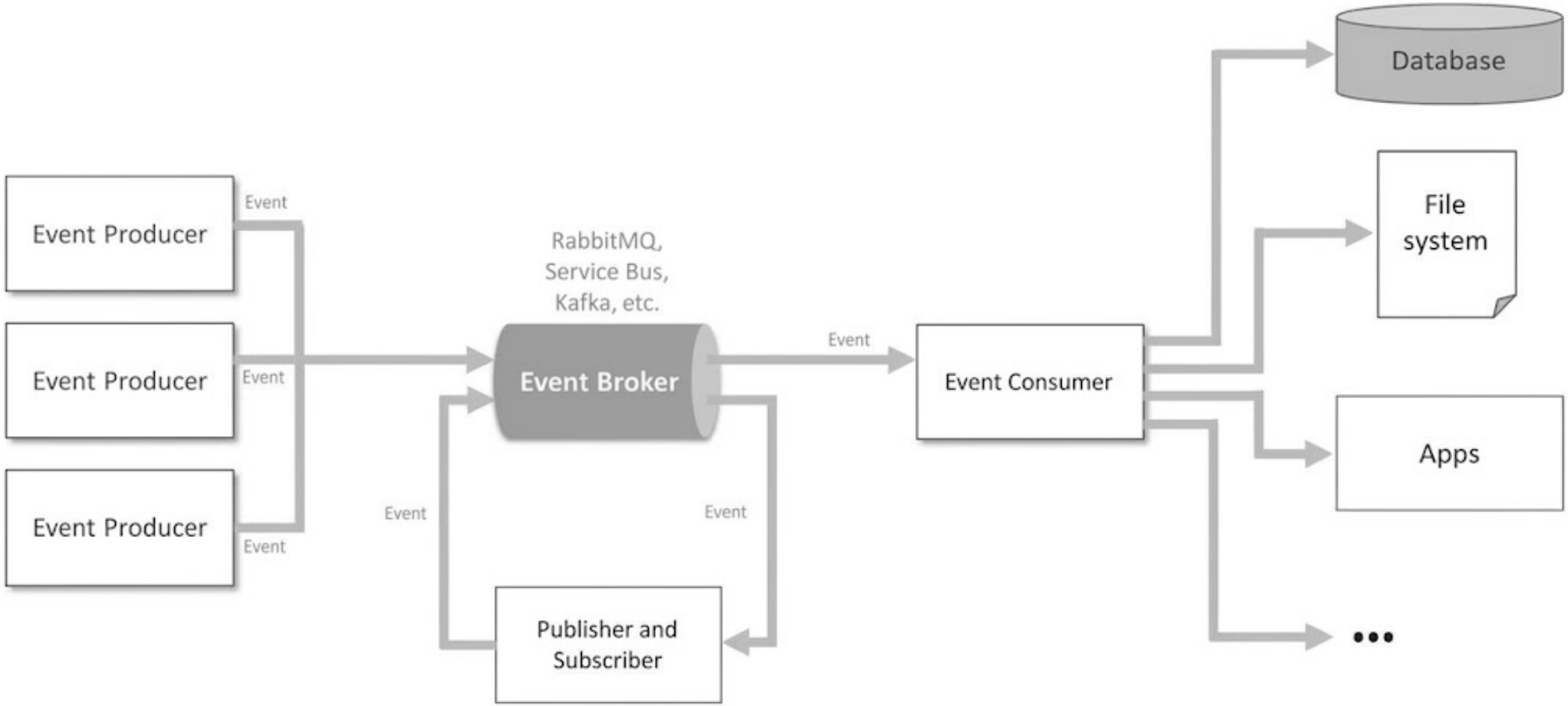
Medallion Architecture

Datawarehousing

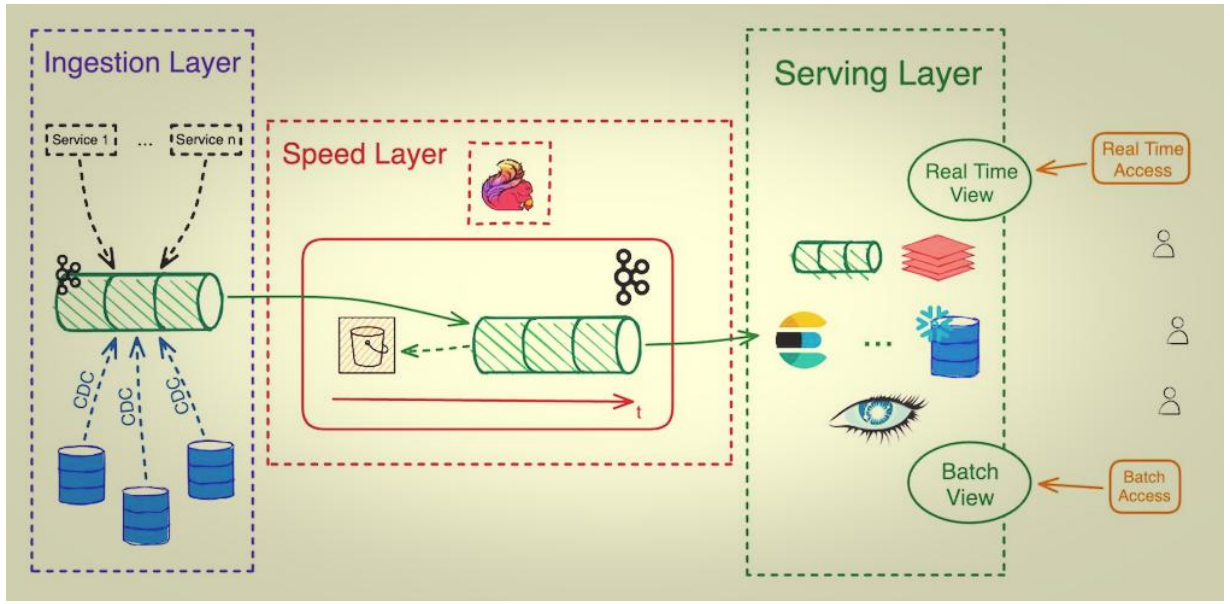


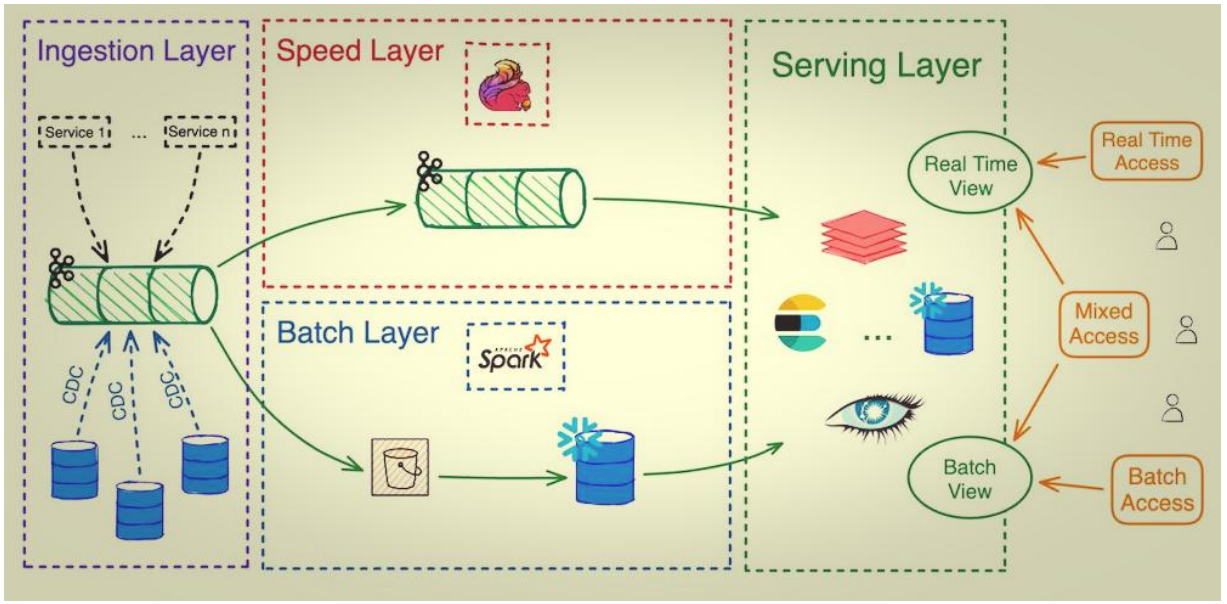
http://www.joakimdalby.dk/HTM/DimensionalModeling.htm

Event Driven Architecture



Datalake (Batch – Stream – Ingest)





Writing Semantic Layer

A **semantic layer** is a business-facing abstraction layer that sits between raw data storage (such as data warehouses, data lakes, or relational databases) and business intelligence (BI) tools, analytics applications, or end-users.

Its primary purpose is to transform complex, highly technical database schemas—complete with foreign keys, cryptic table names, and raw data types—into a unified, standardized business vocabulary (e.g., "Revenue," "Active Customer," "Churn Rate").

Single Source of Truth

Decoupling

Performance Optimization

Access Control

How the Code Works

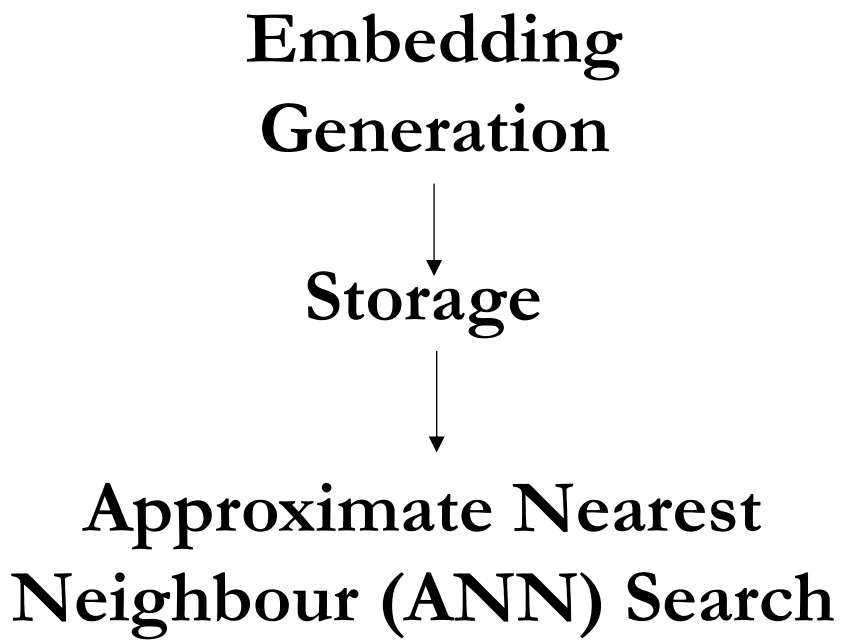
- **Entities & Dimensions:** Define how tables relate to one another (customer_id) and provide categorical or temporal attributes (ordered_at, order_status) for slicing data.
- **Measures:** Define raw aggregations (sum, count, average) directly on database columns.
- **Metrics:** Build business logic on top of measures by applying filters, ratios, or derived calculations (e.g., filtering only for completed orders to calculate total_revenue).
- **Consumption:** Once written, data consumers can query these semantic definitions via SQL, BI connectors, or APIs without needing to write complex JOIN statements or remember table structures.

```
semantic_models:
  - name: fact_orders
    description: "Core transactional order
    data for e-commerce analytics."
    model: ref('stg_orders')

    entities:
      - name: order_id
        type: primary
      - name: customer_id
        type: foreign

    dimensions:
      - name: ordered_at
        type: time
        type_params:
          time_granularity: day
      - name: order_status
        type: categorical

    measures:
      - name: order_amount
        agg: sum
      - name: unique_customers
        agg: count_distinct
        expr: customer_id

metrics:
  - name: total_revenue
    description: "Total gross revenue generated
from completed orders."
    type: simple
    label: "Total Revenue"
    type_params:
      measure: order_amount
    filter: "{{
Dimension('fact_orders__order_status') }} =
'completed'"

  - name: active_customer_count
    description: "Count of unique active
purchasing customers."
    type: simple
    label: "Active Customers"
    type_params:
      measure: unique_customers
```

Vector Databases

A vector database is a specialized data management system designed to store, index, and query high-dimensional vectors.

Unlike traditional databases that query data using exact matches, keywords, or relational filters, a vector database queries data based on semantic similarity.



Distance Metrics: Cosine Similarity, Euclidean Distance, Dot Product

```
SELECT id, content
FROM documents
WHERE category = 'engineering’
ORDER BY embedding <=> '[0.12, -0.43, 0.58, ...]'::vector
LIMIT 5;
```

```
results = index.query(
    vector=[0.12, -0.43, 0.58, ...],
    top_k=5,
    include_metadata=True,
    filter={"category": {"$eq": "engineering"}}
)
```